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Growing Plants

Intended Learning Outcomes

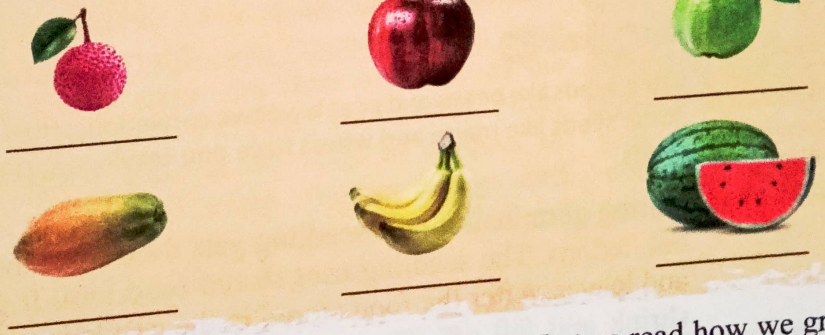
- Growing Plants
- Reproduction by Plant Parts
- Agriculture
- Seed Dispersal
- Growing Plants by Spores
- Stages of Agriculture

Kick Off

Plants are living things. They play a very important role in our lives. They not only give us food, sugar, oil, wood, fibres, rubber, gum, tea, coffee and spices but also give us oxygen for breathing.

Funtopia

Observe the pictures of some fruits given below and write 'no seed', 'one seed', 'few seeds' or 'many seeds' below each picture:

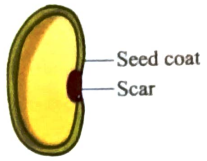


To meet our needs, we need to grow plants. Let us read how we grow plants.

We can grow plants through seeds, stems, roots and leaves.

Reproduction by Seeds

Plants grow from young seedlings. The seedlings gradually grow into a big plant. When these plants grow old they produce new plants and ultimately die. The process by which the plants produce new plants of the same kind is called reproduction.



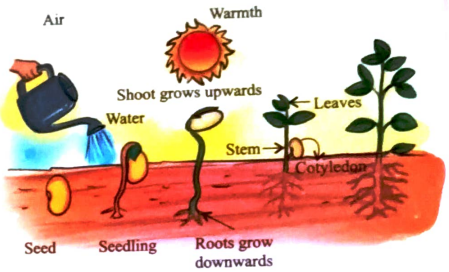
A bean seed



A bean seed splits open

Most of the plants reproduce through seeds. Seeds are present inside the fruits. Some fruits produce one seed and some produce more than one. But all of them do not grow into a new plant. What happens to them?

Some seeds are damaged by heavy rain and strong winds, while some seeds are eaten by insects and birds. Sometimes seeds do not get appropriate soil and adequate water to grow. The seeds that get proper conditions, grow into new plants. The process by which a seed grows and develops into a new plant or seedling is called germination.



Germination of seed

Just a Heads up

Some seeds like beans and peas have two cotyledons; they are called **dicot seeds**. Some seeds like maize and wheat have one cotyledon; they are called **monocot seeds**.

During germination, the seedling gets food from the seed leaves called cotyledons. The seedling uses stored food until it grows its own roots and leaves. After the food stored in the cotyledons has been used, they shrink and fall off. The seedling grows in size and develops leaves which make food for the plant through photosynthesis. The roots of the plants absorb water and minerals from the soil.

Seed Dispersal

If too many seeds are sown too close to each other, they will not be able to germinate. There will be a struggle for food, water and sunlight. Seeds need adequate sunlight, air and water to grow into healthy plants. That is why nature helps in scattering the seeds and ensures that they do not grow too close to each other. The process by which seeds are scattered away from the parent plant is called seed dispersal.

Seeds are dispersed in different ways. They may be carried far away by wind, water or animals. These are called agents of dispersal. Some plants also disperse their seeds explosively. Seed size is an important factor that determines how a seed is dispersed.

Let us understand the methods by which seeds are dispersed.

Dispersal by Wind: Some plants have lightweight seeds and hair-like structure or wings attached to them. These seeds float in the air and are dispersed by the wind. Examples, drumstick seeds, maple seeds, dandelion seeds, etc.



Maple seeds have wing like structures



Dandelion seeds have hair around them



Coconut seed is dispersed by water

Dispersal by Water: Some plants have spongy seeds with fibrous outer coverings, which help these seeds stay afloat in water. Water carries these seeds to large distances. Examples of such plants are lotus (spongy seed) and coconut (fibrous outer covering on the seed).

Dispersal by Animals: Animals usually eat fruits and throw away seeds. Some fruits are eaten up by birds and animals along with seeds. These seeds are then excreted in the droppings or faeces of these animals and birds. Some seeds stick to the body of animals or birds and carried away. Human beings and animals eat fruits like mango, jamun and cherries and throw away their seeds resulting in their dispersal.



Seeds of cocklebur are dispersed by animals

Just a Heads up

In the deserts of North America, elephants eat the fruits that fall from the trees and deposit the seeds in their droppings several miles away.

Dispersal by Bursting/Explosion: Some fruits burst open when they mature. The force of bursting is enough to spread the seeds. Some examples include lady finger (okra), castor, balsam, beans, poppy, pea, etc.



Seeds of beans are dispersed by explosion

Twisting Time

Write T for true and F for false statements.

1. Plants are non-living things.
2. Plants give us nitrogen for breathing.
3. Plants grow from young flowers.
4. Most of the plants reproduce through seeds.
5. Coconut and lotus seeds are dispersed by water.

Reproduction by Plant Parts

Stem, roots and leaves are known as the vegetative parts of the plants. Reproducing by stem, root or leaves of the parent plant is called vegetative reproduction or propagation. Vegetative reproduction is much quicker than reproducing through seeds.

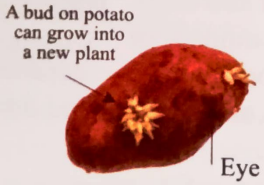
From Stems

Some plants like rose, Hibiscus and money plant can be grown from stem cutting. The plant from which the stem is cut is called the mother plant. Stem cutting is planted in the soil. After a few days, the stem cutting grows into a new plant.



A stem is cut from the mother plant. The stem is planted in the soil;

Growing plant from stem



Potato

Potato, onion, ginger are underground stems. A potato has small buds called eyes on its surface. New plants grow from these buds. Ginger also grows from the buds on the ginger stem. Onion is a bulb-shaped stem. It will grow leaves if you plant one in the ground.

From Roots

Roots of some plants such as sweet potato, carrot, turnip, radish, and beetroot not only store food in them but also produce new plants. These roots are planted in the soil. They are known as tuberous roots.



Sweet potato can grow into a new plant

Place a sweet potato in a glass full of water and keep the glass in an open space. After a few days you will notice a new plant growing out of it.

Activity

This activity shows that some plants reproduce from roots.

You will need: A carrot, shallow dish and water

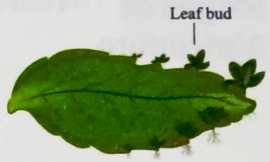
What to do: Cut off the top portion of a carrot that has leaves. Place it in a shallow dish with the bottom part facing downwards. Pour water into the dish and place where it will receive adequate sunlight. Observe after a few days.



You will see: New leaves would have grown from the carrot top.

From Leaves

Some plants reproduce from their leaves. The leaves of Bryophyllum are thick and fleshy, and also have many buds along their edges. When the leaves fall on the soil, these buds get separated and new plants grow from these buds.



Bryophyllum leaf

Growing Plants by Spores



Spores of a fern leaf

Moss and mushrooms are examples of ferns (non flowering plants) and fungi. They do not reproduce by flowers. They produce tiny spores. Each of these spores can grow into a new plant. Fungi, protozoa, algae and mushrooms reproduce through spores.

Agriculture

Plants provide food, oxygen, medicine, timber, etc. to us. We grow plants on a large scale so that we may get enough food and other products to fulfill our needs. This practice of growing plants on a large scale for food or other products is called agriculture.

You must have seen that plants of the same kind are grown in one field. For example, in a particular field you will see only wheat plant or only rice plants. You will never find a field in which all the plants of different kinds are grown. Plants of one kind grown in a particular area at a particular time are called crops.



Crops grown in a farm

The particular area in which a crop is grown is called a farm and the person who works on a farm is known as a farmer.

In India, crops that are grown in the summer season are called *kharif*

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crops. Rice, *jowar*, *bajra*, jute, hemp, peanuts, millet, maize and pulses are examples of *kharif* crops. Vegetables such as spinach, brinjal, tomato, onion, pumpkin, garlic and gourd grow well in summer season. Peach, mango, litchi, watermelon, muskmelon, plum are some fruits that grow best in summer.

Crops that are grown in the winter season are called *rabi* crops. Wheat and gram are *rabi* crops. Radish, turnips, carrots, peas, cauliflower, cabbage and beans are some vegetables that grow in winter season. Apples, oranges and pomegranates are fruits that are generally grown in the winter season.

ges of riculture

There are various stages of agriculture. These are:



The field is ploughed.



Manure is added to the soil.



After this, seeds are sown.



The field is supplied with water. This is called **irrigation**.



To protect the crops, they are sprayed with pesticides.



The crops are then cu and gathered. This i called **harvesting**.

Protection of Crops and Storage of Seeds

We need to protect our crops from:



Scarecrow

- Grazing animals like cows and goats.
- Pests such as locusts, grasshoppers and caterpillars by using insecticides or pesticides.
- Different infections and diseases in crops.
- Big animals can be kept away by proper fencing. After harvesting grains need to be protected against moisture. They also have to be kept safe from rats, moles, birds, squirrels and insects. So, they must be stored in airtight and sealed containers.
- Plants, just like other living things, also need love and care. Under favourable conditions, they grow well. A good crop adds to the wealth of a country.

Make Teaching Fun

We have plants inside our houses too. These indoor plants become members of our family. How will you take care of them?

Words Aid

- **Germination:** the process by which a seed grows into a new plant
- **Cotyledons:** part of the seed that contains food for the baby plant
- **Dispersal:** scattering of seeds away from the parent plant
- **Agriculture:** the science of growing crops
- **Irrigate:** to provide water to the crops
- **Pesticides:** chemicals that are used to kill insects and fungi that damage plants
- **Seedling:** a new plant that has grown from a seed
- **Manure:** plants' and animals' wastes used for enriching the soil with nutrients
- **Kharif crops:** crops that grow in summer season
- **Rabi crop:** crops that grow in winter season

Quick Review

- Plants grow from young seedlings.
- The process by which the plants produce new plants of the same kind is called reproduction.
- Germination is the process by which a seed grows and develops into a new plant or seedling.
- During germination, the seedling gets food from the seed leaves called cotyledons.
- The process by which seeds are scattered away from the parent plant is called seed dispersal.
- Plants depend on wind, water, animals or explosion for seed dispersal.
- Plants can also reproduce through their body parts like stem, root and leaves. This kind of reproduction is called vegetative reproduction.
- Agriculture is the practice of growing plants on a large scale for food or other products.

Put on your Thinking cap

A. Darken the correct option in OMR SHEET :

1. Seeds are present inside :

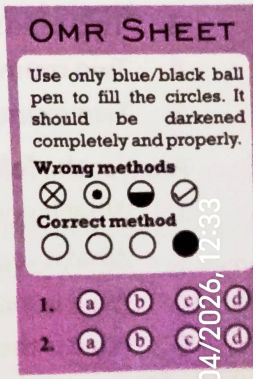
   a. leaves b. fruits

   c. flowers d. roots

2. The process by which a seed grows into a new plant is called :

   a. germination b. dispersal

   c. cotyledon d. agriculture



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3. The roots of the plants absorb water and minerals from the :
   a. leaves b. flowers
   c. root d. soil
4. Which of the following seeds is dispersed by water?
   a. Xanthium b. Coconut
   c. maple d. mango
5. *Jowar* and *bajra* are the crops of :
   a. rabi b. kharif
   c. seasonal d. none of these

3. a b c
4. a b c
5. a b c

B. Fill in the blanks with the correct words from the box :

eyes, spores, fleshy, minerals, farm

1. The leaves of Bryophyllum are thick and fleshy.
2. spores are found on the underside of a fern leaf.
3. The person who works on a farm is known as a farmer.
4. A potato has small buds called eyes.
5. The roots of the plants absorb water and minerals from the soil.

C. Tick (✓) the correct statements and cross (✗) the wrong ones :

1. Plants play a very important role in our lives. ✓
2. The process by which the plants produce new plants is called seedling.
3. Some flowers are damaged by heavy rain and strong wind. ✓
4. The leaves grow in size and develop fruits. ✗
5. Vegetative reproduction is much quicker than reproducing through seeds.

D. Name the following :

1. Two plants that are grown from roots. _____ _____
2. Two plants that are grown from stems. _____ _____
3. Two seeds that are dispersed by explosion. _____ _____
4. Two examples of kharif vegetables. _____ _____
5. Two examples of rabi fruits. _____ _____

E. Answer the following questions :

1. What is germination?
2. What is reproduction?
3. What is the difference between rabi and kharif crops?
4. What is the importance of seed dispersal?

5. How do animals help in seed dispersal?
6. How are cotton and maple seeds dispersed?

Sci-Fun

Number the different stages of germination in the correct order from 1 to 6.



Brain Teaser (HOTS)

- We cannot grow wheat crops in summer (June or July). Why?
- What will happen if you overwater a plant or forget to water it regularly?

Subject Enrichment

Take a container and fill it with soil. Take some sunflower seeds and plant the seeds inside the soil. Place the container somewhere in warm place. Sunlight is good but try to avoid too much direct sunlight, a window sill is a good spot. Keep the soil moist by watering it every day (be careful not to use too much water). Observe them after a week. You will see that tiny roots will start to come out of the sunflower seeds. Take care and after a few days you will notice that it starts to grow into a plant.





Enhance your Life Skills

- **For the growth of plants, farmers add fertilizers to the plants. Try to know about the constituents of fertilizers and how they are helpful in the growth of plants.**
- **Why is planting trees important for a city?**

Subject Connect

Geography

Agriculture has changed the way of living. Collect information about different types of crops grown in different parts of India such as in Maharashtra, Kerala, Himachal Pradesh and Assam. On a map of India mark these states and the crops that grow best in the particular state.